PROJECT 23: FORAGING BEHAVIOUR MODEL

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Problem Statement

This project models an animal searching for food in an environment containing two food patches. Food collected from a patch reduces its available resources, while food gradually regenerates. Moving between patches has an energy cost. The objective is to learn a policy that maximizes the animal's cumulative reward over a fixed number of time steps.

State, Action, Reward and Environment

RL component	Project definition
Agent	A simulated animal that makes sequential foraging decisions.
State (S)	A tuple containing the animal's current location (Patch A or B), the remaining food level in Patch A, and the remaining food level in Patch B. Food levels are represented as discrete values.
Action (A)	Two possible actions: (1) forage at the current patch, or (2) move to the other patch.
Reward (R)	When foraging, the agent receives reward equal to the food collected (up to 2 units per step). When moving, it receives a penalty of −0.5 to represent energy expenditure.
Environment	A simulated two-patch world. Patch A has a maximum capacity of 10 food units and Patch B has 8. Food regenerates by 1 unit per patch after each action, without exceeding the patch's capacity. Each episode lasts 60 time steps.

Learning Algorithm and Objective

The project uses tabular Q-learning with an epsilon-greedy action-selection policy. The agent explores actions with probability ε and otherwise chooses the action with the highest estimated Q-value. The Q-value update is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

The learning rate is α = 0.15, the discount factor is γ = 0.95, and the initial exploration probability is ε = 0.20 with decay during training. The agent is trained for 1,200 episodes. Its performance is evaluated against a random policy and a fixed heuristic policy.

Evaluation

The main evaluation measure is cumulative reward per episode. Learning curves use 40-episode moving averages to compare Q-learning with the random and heuristic baselines. The goal is to determine whether the agent learns a useful food-collection and movement strategy.